

JESSE WANG

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EDUCATION

University of Massachusetts Amherst

Amherst, MA

B.S. in Computer Science and Mathematics

Expected Date of Graduation: May 2028

Relevant Coursework: Data Structures, Multivariable Calculus, Discrete Mathematics, Differential Equations

Glastonbury High School

Glastonbury, CT

High Honors — GPA: 3.92/4.00

August 2022 - June 2026

Awards: 3rd @ MIT BluePrint, Amazon Future Engineer Scholar

Relevant Coursework: Computer Programming in C++, Data Structures & Algorithms, AP Statistics

EXPERIENCES

Amazon

Seattle, WA

Software Engineering Intern

Incoming Summer 2027

FIRST Robotics

August 2022 - May 2026

Team 2170 Programming & Electronics Captain

May 2025 - May 2026

- Led FRC Team 2170 as captain, expanding our team from **28 to 63 students**, welcoming and training new members
- Designed and taught **Java** lessons, preparing students for real development, managed through **GitHub Code Reviews**
- Traveled door-to-door to **500+ local businesses**, securing over **\$6000 in sponsorships** and forming connections with team partners

Vision & Scouting Lead

August 2023 - May 2025

- Developed real-time computer vision system using **YOLOv8/YOLOv11 neural networks** for FIRST Robotics Competition, achieving **85% accuracy** with **<25ms inference latency** for autonomous robot operation
- Optimized deep learning models for edge deployment by implementing **INT8 quantization**, reducing model size by **73% (11MB→2.9MB)** and compiling for **Google Coral Edge TPU** hardware acceleration
- Engineered complete ML pipeline from dataset collection through deployment, including transfer learning from **COCO weights**, GPU training on **Google Colab**, and multi-format export for **TensorFlow, PyTorch, ONNX, and TFLite**

MIT Introduction to Technology, Engineering, and Science (MITES) Semester

June 2025 - December 2025

Scholar

- Received the **Best of MITES award** for my writing on the limitations of AI
- Received the **MITES Semester Achievement Award** for PulseLine, a 24/7 **agentic AI** medical assistant capable of autonomously handling patient phone calls and online requests, including scheduling, prescription refills, and doctor lookups
- Real-time voice transcription and call routing pipeline built with **Twilio** and **OpenAI Realtime API** with **WebSocket** communication
- Designed a **FastAPI** backend connected to a **PostgreSQL** patient database to enable personalized, context-aware responses

RESEARCH

MMIST-ccRCC Benchmark Optimization

December 2025 - Present

Columbia University

New York, NY

- Creating a cancer assessment model targeting **Clear Cell Renal Cell Carcinoma (ccRCC)** with Professor Adam Lin
- Investigating solutions to critical performance gaps identified in the **MMIST-ccRCC** benchmark by replacing traditional task-specific encoders (such as 3D-ResNet18) with Google's state-of-the-art **CT Foundation Model**

Automated Brain Tumor Segmentation & Progression Prediction

August 2024 - June 2026

University of Connecticut

Storrs, CT

- Developed an ML pipeline for automated brain tumor segmentation & progression prediction, achieving a **mean IoU score of 0.5**
- Created a Python pipeline to tokenize image masks from converted MRI files via a **VQGAN model**
- Fine-tuned **Meta AI's Segment Anything Model 2** on the Cancer Imaging Archive's **Brain Tumor Progression** dataset

RELEVANT PROJECTS

Chess Engine

Java, JavaFX

- A full-featured chess engine implementing **alpha-beta search** with **Zobrist hashing**, **transposition tables**, **null-move pruning**, and **late-move reductions** to improve search efficiency
- Developed an interactive JavaFX GUI with **move validation**, move highlighting, **game state management**, and takeback functionality

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (PostgreSQL), JavaScript, HTML/CSS, Kotlin

Developer Tools: Git, Visual Studio Code, PyCharm, IntelliJ, Ngrok, Jupyter Notebook, Anaconda, Arduino, OpenAI API, Groq, Twilio

Libraries: pandas, NumPy, Matplotlib, OpenCV, scikit-learn, Hugging Face Transformers, Ultralytics

Frameworks: FastAPI, WPIlib, PyTorch, TensorFlow, Keras, Django